

The pattern of 1s and 0s in Table 4–4 is called a *thermometer code* because the number of 1s increases monotonically as v_s increases, like the way the mercury column in a thermometer increases with temperature. Integrated circuit flash converters have built-in decoders that convert the thermometer code into a standard binary code. The flash converter is by far the fastest ADC architecture.

Exercise 4–40

The reference voltage in Figure 4–84(a) is $V_{\text{REF}} = 15 \text{ V}$. What are the output codes corresponding to $v_s = 1, 2, 5, 10$, and 14 V ?

Answers: (0, 0, 0, 0); (0, 0, 0, 0); (0, 0, 0, 1); (0, 1, 1, 1); (1, 1, 1, 1)

SUMMARY

- A linear dependent source generates a voltage or a current whose value is proportional to a voltage or current at another point in a circuit. There are four such sources: the current-controlled voltage source, the voltage-controlled voltage source, the current-controlled current source, and the voltage-controlled current source.
- Circuits containing dependent sources can be analyzed by node-voltage or mesh-current methods. Such circuits can have input-output relationships that produce voltage, current, or power gain. The presence of feedback can dramatically influence the input and output resistances of active circuits.
- The OP AMP is an active device with five terminals called the inverting input, the noninverting input, the output, and two power supply terminals. The device is a high-gain differential amplifier with three possible operating modes: +saturation, –saturation, and linear. The output predicted by the linear mode circuit model is compared with known bounds to determine the actual operating mode.
- The ideal OP AMP model has an infinite voltage gain, an infinite input resistance, and zero output resistance. The i – v characteristics of an ideal OP AMP are $i_P = i_N = 0$ and $v_P = v_N$. The ideal model is a good working approximation in linear applications.
- The four basic OP AMP circuit building blocks are the inverting amplifier, the noninverting amplifier, the inverting summer, and the subtractor. The analysis or design of complex OP AMP circuits can be based on these four building blocks provided the interconnections are made between the output of one to the input of another.
- Important applications of OP AMPs include digital-to-analog converters, instrumentation systems, and comparator circuits.

PROBLEMS

OBJECTIVE 4–1 LINEAR ACTIVE CIRCUITS (SECTS. 4–1, 4–2)

Given a linear resistance circuit containing dependent sources, find selected output signals, input-output relationships, or input-output resistances.

See Examples 4–1, 4–2, 4–3, 4–4, 4–5, 4–6, 4–7, 4–8, 4–9, 4–10, 4–11 and Exercises 4–1, 4–2, 4–3, 4–5, 4–6, 4–7, 4–8, 4–9, 4–10, 4–11, 4–12, 4–13.

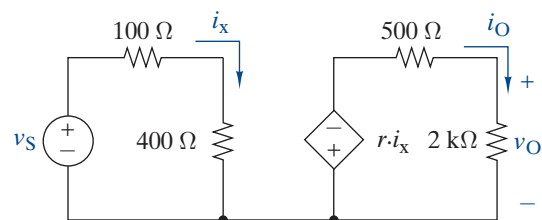


FIGURE P4–1

4–1 Find the voltage gain v_o/v_s and current gain i_o/i_x in Figure P4–1 for $r = 5 \text{ k}\Omega$.

4–2 Find the voltage gain v_o/v_1 and the current gain i_o/i_s in Figure P4–2. For $i_s = 5 \text{ mA}$, find the power supplied by the

input current source and the power delivered to the 2-k Ω load resistor.

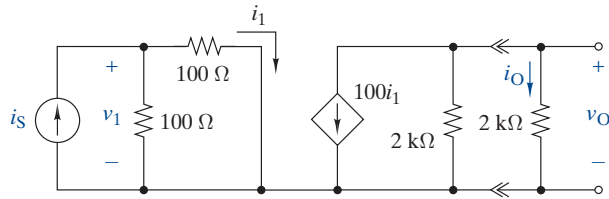


FIGURE P4-2

4-3 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-3 for $g = 2 \times 10^{-3}$ S. For $v_S = 5$ V, find the power supplied by the input voltage source and the power delivered to the 2-k Ω load resistor.

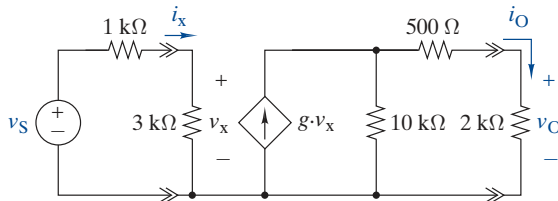


FIGURE P4-3

4-4 (a) Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-4.
(b) Validate your answers by simulating the circuit in OrCAD.

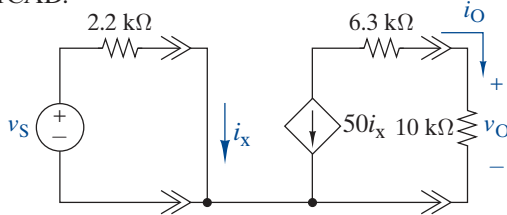


FIGURE P4-4

4-5 Find the voltage gain v_O/v_S in Figure P4-5.

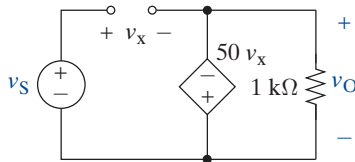


FIGURE P4-5

4-6 Find an expression for the current gain i_O/i_S in Figure P4-6.
Hint: Apply KCL at node A.

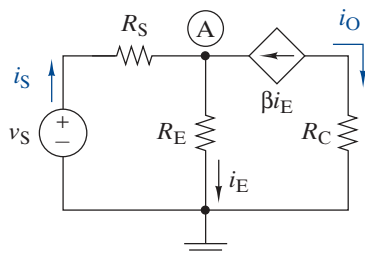


FIGURE P4-6

4-7 (a) Find the voltage v_O in Figure P4-7.
(b) Validate your answer by simulating the circuit in OrCAD.

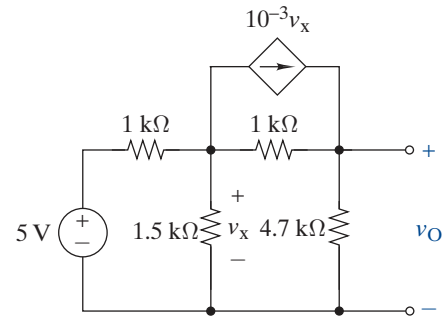


FIGURE P4-7

4-8 (a) Find an expression for the gain i_O/v_S in Figure P4-8 in terms of R_X .
(b) Select a value for R_X so that the gain is -0.227 .

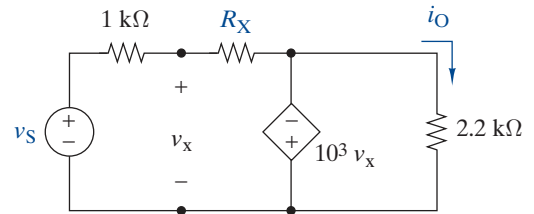


FIGURE P4-8

4-9 Find an expression for the voltage gain v_O/v_S in Figure P4-9.

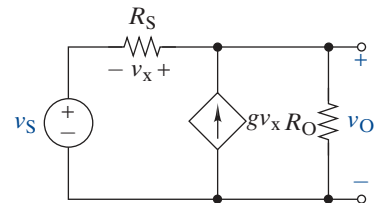


FIGURE P4-9

4-10 (a) Find an expression for the voltage gain v_O/v_S in Figure P4-10.
(b) Let $R_S = 10$ k Ω , $R_L = 10$ k Ω and $\mu = 100$. Find the voltage gain v_O/v_S as a function of R_F . What is the voltage gain when R_F is an open circuit, a short circuit, and for $R_F = 100$ Ω ?
(c) Simulate the circuit in OrCAD by varying R_F from 1 Ω to 10 M Ω . Read your output $R_F = 100$ Ω . How does your answer compare with part (b)?

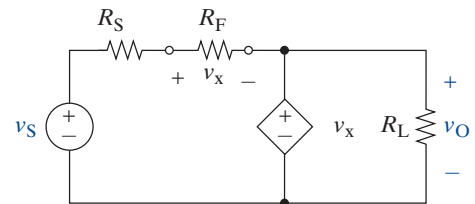


FIGURE P4-10

- 4-11 Select g in the circuit of Figure P4-11 so that the output voltage is 10 V.

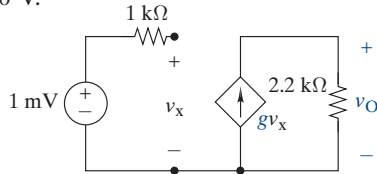


FIGURE P4-11

- 4-12 **D** Design a dependent source circuit that has a closed-loop voltage gain of -10 using a VCVS with a μ of 100. The source circuit is a voltage source v_S in series with a 1-k Ω resistor, and the load is a 3.3-k Ω resistor. (*Hint:* See Figure P4-10.)

- 4-13 Find the Thévenin equivalent circuit that the load R_L sees in Figure P4-13. Repeat the problem with R_F replaced by an open circuit.

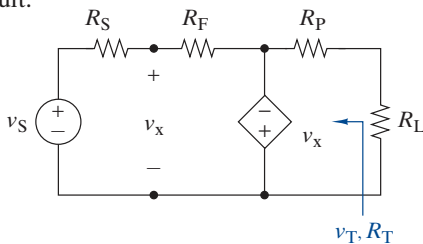


FIGURE P4-13

- 4-14 Find the Thévenin equivalent circuit that the load R_L sees in Figure P4-14.

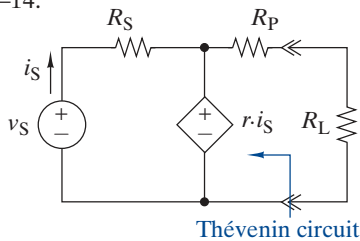


FIGURE P4-14

- 4-15 Find R_{IN} in Figure P4-15

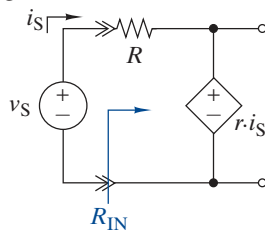


FIGURE P4-15

- 4-16 Find R_{IN} in Figure P4-16.

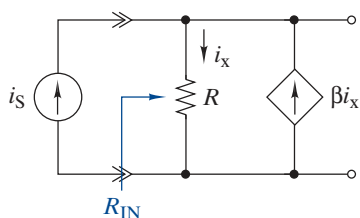


FIGURE P4-16

- 4-17 Find the Norton Equivalent circuit seen by the load in Figure P4-17.

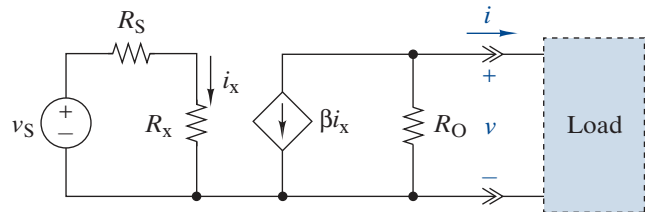


FIGURE P4-17

- 4-18 Find the Thévenin Equivalent circuit seen by the load in Figure P4-18.

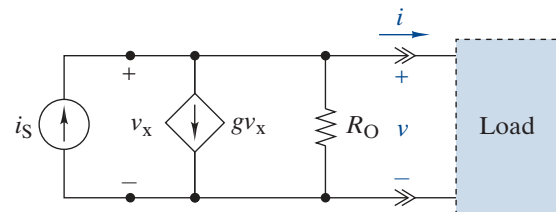


FIGURE P4-18

- 4-19 The circuit parameters in Figure P4-19 are $R_B = 100$ k Ω , $R_C = 3.3$ k Ω , $\beta = 100$, $V_\gamma = 0.7$ V, and $V_{CC} = 15$ V. Find i_C and v_{CE} for $v_S = 1$ V. Repeat for $v_S = 5$ V.

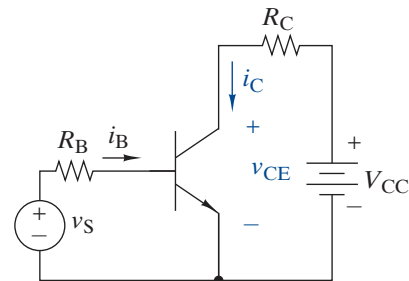


FIGURE P4-19

- 4-20 **D** The circuit parameters in Figure P4-19 are $R_C = 3$ k Ω , $\beta = 120$, $V_\gamma = 0.7$ V, and $V_{CC} = 5$ V. Select a value of R_B such that the transistor is in the saturation mode when $v_S \geq 2$ V.

- 4-21 The parameters of the transistor in Figure P4-21 are $\beta = 60$ and $V_\gamma = 0.7$ V. Find i_C and v_{CE} for $v_S = 0.8$ V. Repeat for $v_S = 2$ V.

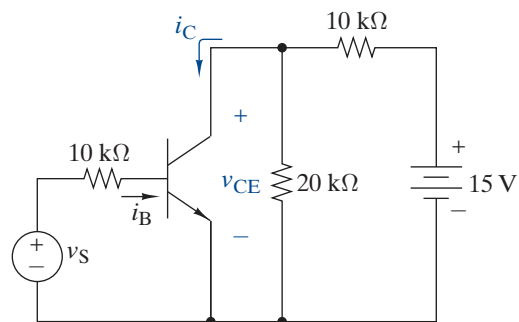


FIGURE P4-21

- 4-22 **D** An emergency indicator light uses a 10 V, 2-W incandescent lamp. It is to be ON when a digital output is high (5 V). The digital circuit does not have sufficient power to turn on the lamp directly. However, as is common practice, a transistor driver is used as a digital switch. Select R_B in the circuit of Figure P4-22 so to drive the transistor into saturation causing it to act as a short circuit between the lamp and ground when the digital output is high. The Thévenin equivalent for the digital circuit is also shown in the figure.

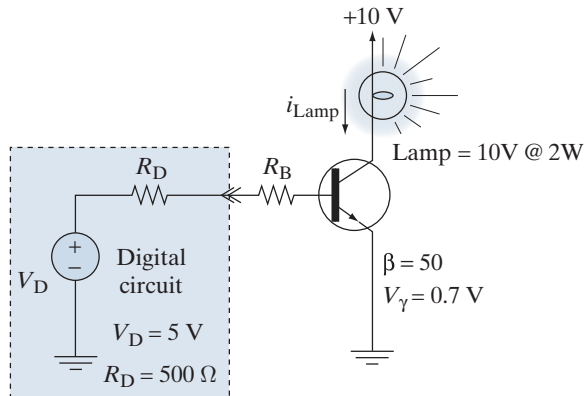


FIGURE P4-22

OBJECTIVE 4-2 OP AMP CIRCUIT ANALYSIS (SECTS. 4-3, 4-4)

Given a linear resistance circuit containing OP AMPs, find selected output signals or input-output relationships. See Examples 4-12, 4-13, 4-14, 4-15, 4-16, 4-17, 4-18, 4-19, 4-20 and Exercises 4-14, 4-15, 4-16, 4-17, 4-18, 4-19, 4-20, 4-21, 4-22, 4-23, 4-24, 4-25, 4-26, 4-27, 4-28.

- 4-23 Find the voltage gain of each OP AMP circuit shown in Figure P4-23.

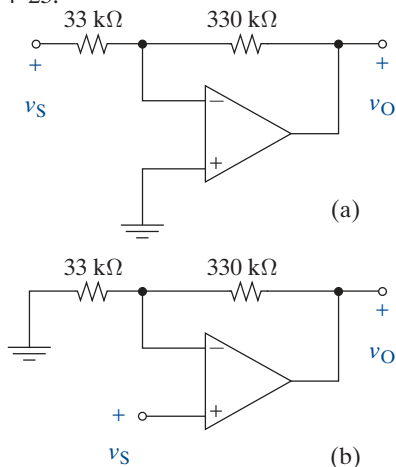


FIGURE P4-23

- 4-24 **D** Considering simplicity and standard 10% tolerance resistors as major constraints, design OP AMP circuits that produce the following voltage gains $\pm 10\%$: -100 , $+200$, $+1$, -0.5 , $+0.5$.

- 4-25 **E** Two OP AMP circuits are shown in Figure P4-25. Both claim to produce a gain of either ± 100 .

(a) Show that the claim is true.

(b) A practical source with a series resistor of 1 kΩ is connected to the input of each circuit. Does the original claim still hold? If it does not, explain why?

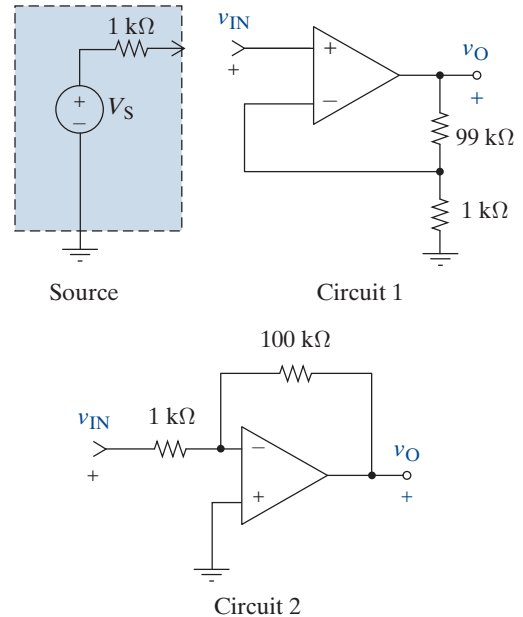


FIGURE P4-25

- 4-26 **E** Suppose the output of the practical source shown in Figure P4-25 needs to be amplified by -10^4 and you can use only the two circuits shown. How would you connect the circuits to achieve this? Explain why.

- 4-27 **E** (a) Find the voltage gain v_O/v_S in Figure P4-27.

(b) Validate your answer by simulating the circuit in OrCAD.

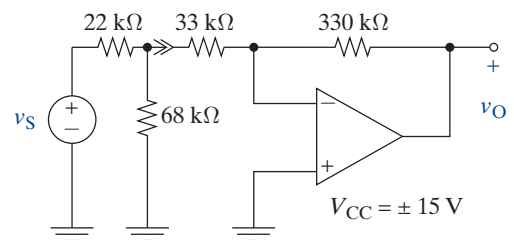


FIGURE P4-27

- 4-28 What is the range of the gain v_O/v_S in Figure P4-28?

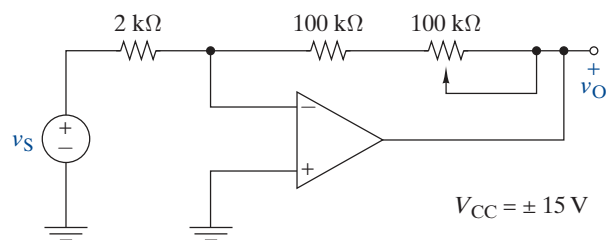


FIGURE P4-28

4-29 **D** Design a simple OP AMP circuit that has a variable gain from +10 to +100.

4-30 For the circuit in Figure P4-30:

- (a) Find v_O in terms v_S .
 (b) Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

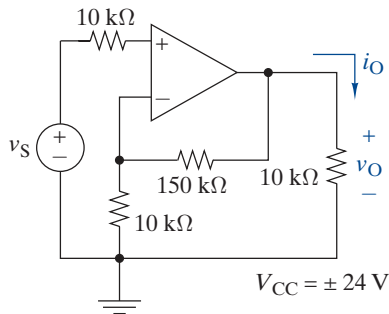


FIGURE P4-30

4-31 For the circuit in Figure P4-31:

- (a) Find v_O in terms v_S .
 (b) Find i_O for $v_S = 0.5$ V. Repeat for $v_S = 2$ V.

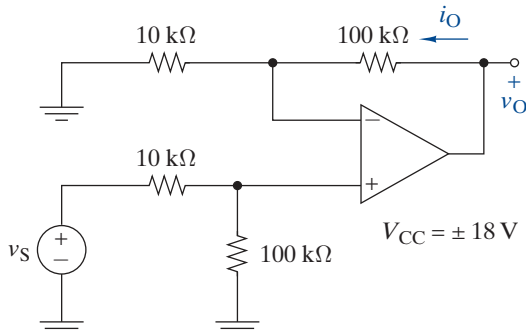


FIGURE P4-31

4-32 **E** A young designer needed to amplify a 2-V signal by the factors of 1, 5, and 10. Find the problem with the design shown in Figure P4-32. Recommend a fix.

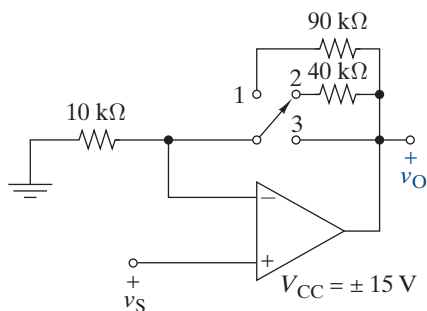


FIGURE P4-32

4-33 **D** Design two circuits to produce the following output:
 $v_O = 2v_1 - 3v_2$.

- (a) In your first design use a standard subtractor.
 (b) In your second design both inputs must be into high input resistance amplifiers to avoid loading.

4-34 For the circuit in Figure P4-34:

- (a) Find v_O in terms of the inputs v_1 and v_2 .
 (b) If $v_1 = 1$ V, what is the range of values v_2 can have without saturating the OP AMP?

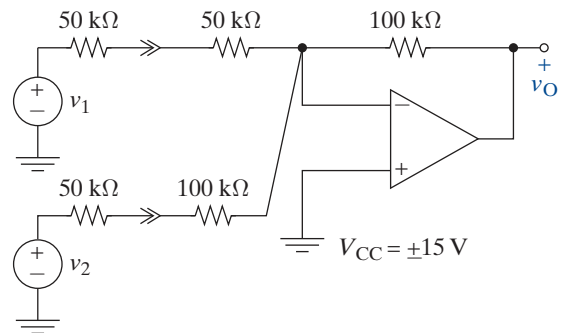


FIGURE P4-34

4-35 The input-output relationship for a three-input inverting summer is

$$v_O = -[v_1 + 10v_2 + 100v_3]$$

The resistance of the feedback resistor is 100 kΩ. Find the values of the input resistors R_1 , R_2 , and R_3 .

4-36 Find v_O in terms of the inputs v_1 and v_2 in Figure P4-36.

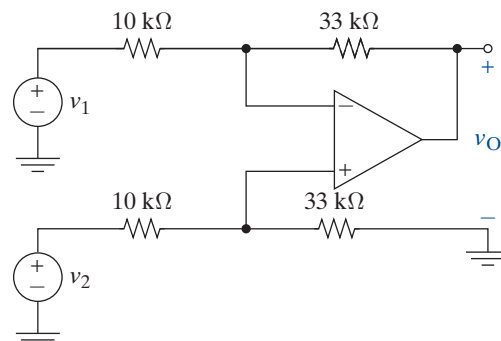


FIGURE P4-36

4-37 The switch in Figure P4-37 is open, find v_O in terms of the inputs v_{S1} and v_{S2} . Repeat with the switch closed.

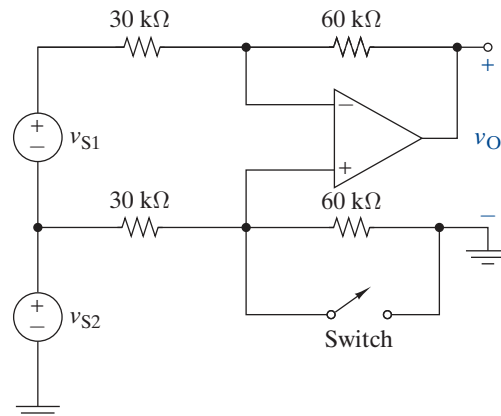


FIGURE P4-37

4-38 Find v_O in terms of v_{S1} and v_{S2} in Figure P4-38.

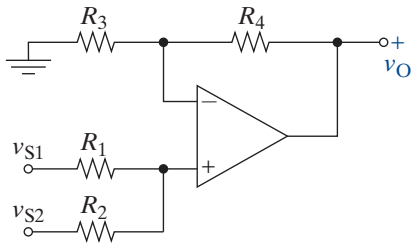


FIGURE P4-38

4-39 It is claimed that $v_O = v_S$ when the switch is closed in Figure P4-39 and that $v_O = -v_S$ when the switch is open. Prove or disprove this claim.

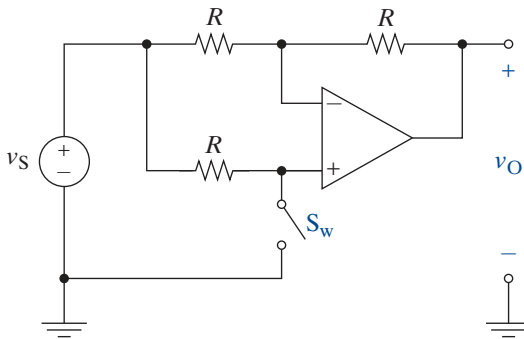


FIGURE P4-39

4-40 The circuit in Figure P4-40 has a diode in its feedback path and is called a “log-amp” because its output is proportional to the natural log of the input.

- (a) Show that $v_O = -V_T \ln(1 + v_S/(R_S I_O))$ if the i - v characteristics of the diode is $i_D = I_O(e^{v_D/V_T} - 1)$.
 (b) Using MATLAB plot v_O versus v_S for $R_S = 10 \text{ k}\Omega$, $I_O = 2 \times 10^{-14} \text{ A}$ and $V_T = 0.026 \text{ V}$. Plot your results on a semilog plot for $10^{-6} \text{ V} \leq v_S \leq 100 \text{ V}$.

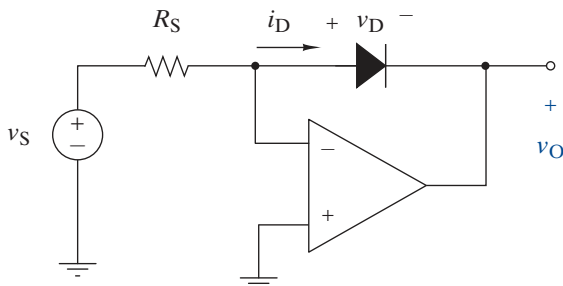


FIGURE P4-40

4-41 (a) Use node voltage analysis to find the input-output relationship or K of the circuit in Figure P4-41.

(b) Select values for the resistors so that $K = -6$.

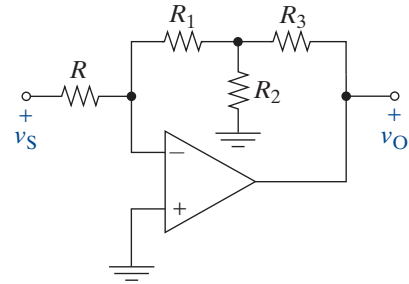


FIGURE P4-41

4-42 Use node voltage analysis in Figure P4-42 to show that $i_O = -v_S/2R$ regardless of the load. That is, show that the circuit is a voltage-controlled current source.

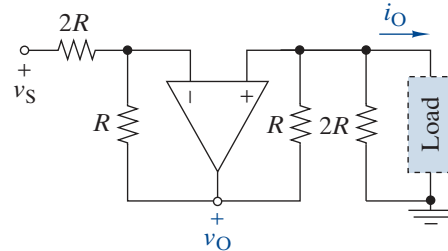


FIGURE P4-42

4-43 For the circuit of Figure P4-43:

- (a) Find the output in terms of v_1 and v_2 .
 (b) Draw a block diagram for the circuit.

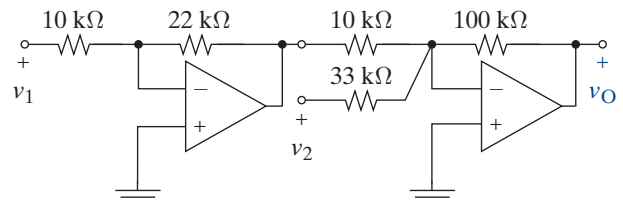


FIGURE P4-43

4-44 Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4-44.

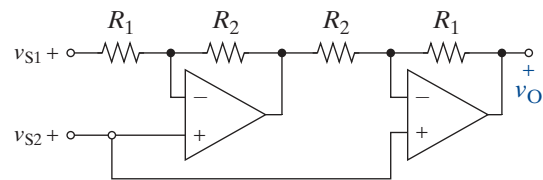


FIGURE P4-44

4-45 For the block diagram of Figure P4-45:

- (a) Find an expression for v_O in terms of v_1 and v_2 .
 (b) Design a suitable circuit that realizes the block diagram using only one OP AMP.

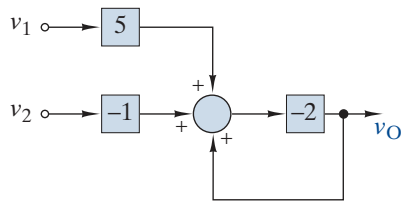


FIGURE P4-45

4-46 For the block diagram of Figure P4-46:

- Find an expression for v_O in terms of v_S and the input voltage source.
- Design a suitable circuit that realizes the block diagram using only one OP AMP and the 0.5-V source.

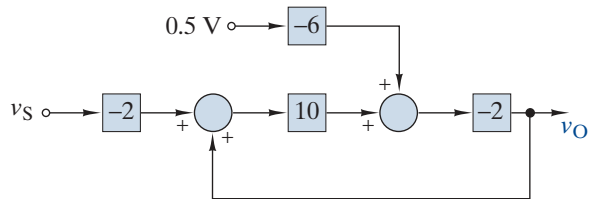


FIGURE P4-46

4-47 For the circuit in Figure P4-47

- Find v_O in terms of v_S and the 1-V source.
- Prove that the block diagram provides the same output.
- Redesign the circuit using only one OP AMP.
- Validate your design using OrCAD.

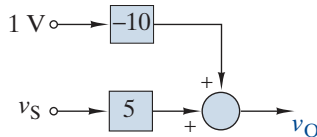
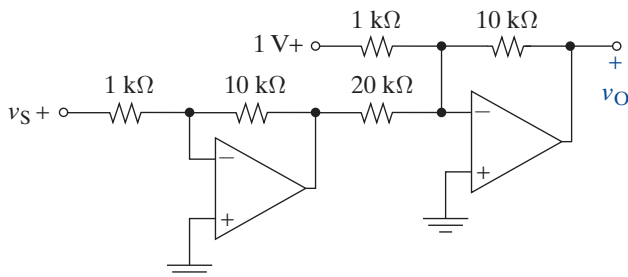
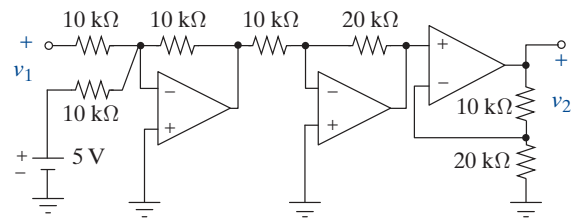
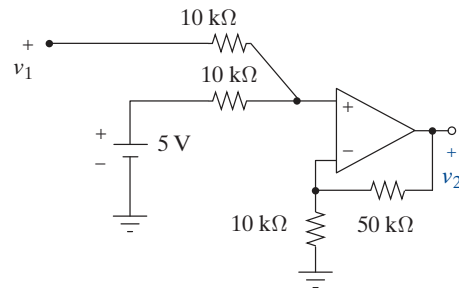


FIGURE P4-47

4-48 On an exam, students were asked to design an efficient solution for the following relationship: $v_2 = 3v_1 + 15$. Two of the designs are shown in Figure P4-48. Which, if any, of the designs are correct and what grade would you award each student?



(a)



(b)

FIGURE P4-48

4-49 For the circuit of Figure P4-49:

- Find the output v_2 in terms of the input v_1 in Figure P4-49.
- Draw a representative block diagram for the circuit.

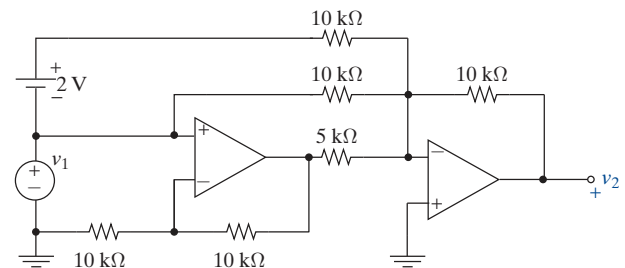


FIGURE P4-49

4-50 For the circuit of Figure P4-50:

- Use node-voltage analysis to find the output v_O in terms of the input v_S .
- Draw a representative block diagram for the circuit.
- Verify your answer using OrCAD.

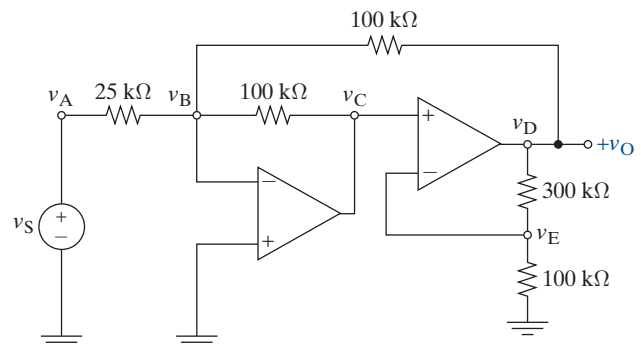


FIGURE P4-50